



**DEPARTMENT OF CHEMICAL ENGINEERING**  
**NATIONAL INSTITUTE OF TECHNOLOGY SRINAGAR**

**Bioresource Technology (CET-465)**

**Home Assignment plus Question Bank-5**

| S. No. | Questions                                                                                                                                              | COs |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1.     | Write a brief note on activated carbon.                                                                                                                | CO2 |
| 2.     | Why is there need for switching over from traditional feedstock (fossil origin) to the bioresources for production of activated carbon?                | CO1 |
| 3.     | Discuss in short the various biomass feedstock for activated carbon.                                                                                   | CO1 |
| 4.     | What are different types of activated carbon? How is the structure explained?                                                                          | CO2 |
| 5.     | Briefly summarize the properties of activated carbon. How are the pores so important, explain with the help of a suitable diagram?                     | CO2 |
| 6.     | Describe the process for production of activated carbon with the help of a suitable flow diagram.                                                      | CO3 |
| 7.     | Depict various possible routes for synthesis of active carbon from biomass with the help of a diagram.                                                 | CO3 |
| 8.     | What are different applications of activated carbon? Discuss the factors affecting production.                                                         | CO2 |
| 9.     | Write a brief note on biofuels. Discuss the relevance of biofuel technology for developing countries like India.                                       | CO3 |
| 10.    | Discuss the significance of Fischer–Tropsch (FT) synthesis technology for biofuel production.                                                          | CO3 |
| 11.    | Briefly discuss the global scenario with respect to biofuel production.                                                                                | CO2 |
| 12.    | Discuss at least five key biomass feedstock for biofuel production.                                                                                    | CO1 |
| 13.    | How are the biofuels classified on the basis of feedstock, discuss them?                                                                               | CO1 |
| 14.    | What are various key issues and performance considerations with respect to biofuel sustainability?                                                     | CO1 |
| 15.    | Write a short note on biodiesel along with its advantages and disadvantages.                                                                           | CO1 |
| 16.    | Enlist the raw materials used for production of biodiesel. Discuss at least five of them.                                                              | CO1 |
| 17.    | Explain the chemical reaction involved in formation of triglycerides. Enlist at least ten fatty acids present in fats used in formation of biodiesel.  | CO2 |
| 18.    | Describe the chemistry of esterification and transesterification along with its importance in production of biodiesel.                                 | CO2 |
| 19.    | Discuss the commercial production of biodiesel with the help of a flow chart. Also explain the chemical reactions involved.                            | CO3 |
| 20.    | Tabulate the standard properties of biodiesel in the light of American and Indian standards.                                                           | CO2 |
| 21.    | Make a comparison between biodiesel and petrodiesel with respect to their physical properties and emission.                                            | CO2 |
| 22.    | Explain the terms like B5, B15, B100. Take care of the questions and answers given in the Lecture-16 (biodiesel).                                      |     |
| CO1    | Fundamental understanding of the bioresources and its applications for attainment of social objectives (energy, environment, product, sustainability). |     |
| CO2    | Acquire knowledge with respect to the properties of the bioresources and the conversion technologies.                                                  |     |
| CO3    | Exhibiting knowledge of the systems used for bioresources and bioresource technology.                                                                  |     |
| CO4    | Understanding about analysis of data and their applications in design of the systems and development of the bioprocess.                                |     |